

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Subject: 392132 Physics II

Date: 25 May 2015

Year: 2014

Section: 15 - 16

Time: 13.00-15.00

Name _____ ID _____ Class _____

Instructions:

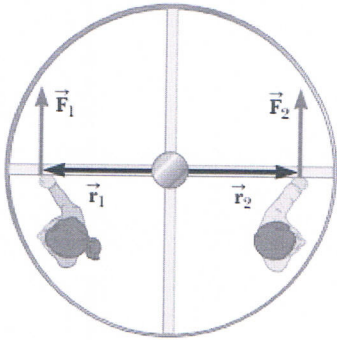
1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks are **NOT** allowed.
4. Express your answer in English. Other languages will be discarded.
5. A scientific calculator is **NOT** allowed.
6. Electronic communication device are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page), 12 questions and a total score of 50 points.
8. Write all your answers on this exam sheet.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

1. A model rocket is constructed with a motor that can provide a total impulse of $40 \text{ N}\cdot\text{m}$. The mass of the rocket is 0.2 kg. What is the speed that this rocket achieve when launched from rest? Neglect the effects of gravity and air resistance. (2 points)
2. A 100 g tennis ball is traveling straight at a player at 20.0 m/s. The player volleys the ball straight back at 40.0 m/s. If the ball remains in contact with the racket for 0.05 s, what average force acts on the ball? (3 points)
3. Show that the kinetic energy of a particle of mass m is related to the magnitude of the momentum p of that particle by $KE = P^2/2m$. Note: This expression is invalid for particles traveling at speeds near that of light. (3 points)
4. High-speed stroboscopic photographs show that the head of a 200 g golf club is traveling at 50 m/s just before it strikes a 20 g golf ball at rest on a tee. After the collision, the club head travels (in the same direction) at 40 m/s. Find the speed of the golf ball just after impact. (2 points)

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7. Two disgruntled business people are trying to use a revolving door. The woman on the left exerts a force of 400 N perpendicular to the door and 1.0 m from the hub's center, while the man on the right exerts a force of 600 N perpendicular to the door and 0.400 m from the hub's center. Find the net torque on the revolving door. (3 points)



8. A large grinding wheel in the shape of a solid cylinder of radius 0.5 m is free to rotate on a frictionless, vertical axle. A constant tangential force of 500 N applied to its edge causes the wheel to have an angular acceleration of 0.1 rad/s^2 . $\left(I = \frac{1}{2}mr^2 \right)$

- (a) What is the moment of inertia of the wheel? (2 points)
- (b) What is the mass of the wheel? (2 points)
- (c) If the wheel starts from rest, what is its angular velocity after 10.00 s have elapsed, assuming the force is acting during that time? (2 points)

9. A solid, uniform disk of radius 0.1 m and mass 50.0 kg rolls down a ramp of length 20 m that makes an angle of 30° with the horizontal. The disk starts from rest from the top of the ramp. Find

$$\left(I = \frac{1}{2}mr^2 \right)$$

(a) the speed of the disk's center of mass when it reaches the bottom of the ramp. (5 points)

(b) the angular speed of the disk at the bottom of the ramp. (2 points)

10. Each of the following objects has a radius of 0.1 m and a mass of 2.0 kg, and each rotates about an axis through its center with an angular speed of 10.0 rad/s. Find the magnitude of the angular momentum of each object.

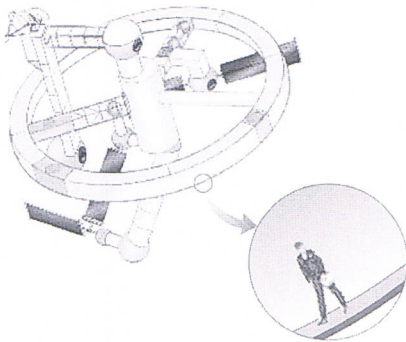
(a) a hoop $(I = mr^2)$ (2 points)

(b) a solid cylinder $\left(I = \frac{1}{2}mr^2 \right)$ (2 points)

(c) a solid sphere $\left(I = \frac{2}{5}mr^2 \right)$ (2 points)

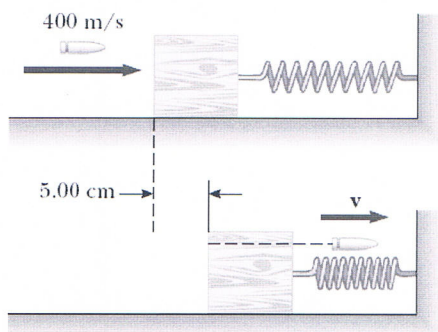
11. A solid, horizontal cylinder of mass 10.0 kg and radius 1.00 m rotates with an angular speed of 2.00 rad/s about a fixed vertical axis through its center. A 0.2 kg piece of putty is dropped vertically onto the cylinder at a point 0.20 m from the center of rotation and sticks to the cylinder. Determine the final angular speed of the system. (4 points) $\left(I = \frac{1}{2}mr^2 \right)$

12. A space station shaped like a giant wheel has a radius of 100 m and a moment of inertia of $5.00 \times 10^8 \text{ kg} \cdot \text{m}^2$. A crew of 150 are living on the rim, and the station's rotation causes the crew to experience an acceleration of 1 g. When 100 people move to the center of the station for a union meeting, the angular speed changes. What acceleration is experienced by the managers remaining at the rim? Assume the average mass of each inhabitant is 50 kg (hint: $a_c = \omega^2 r$) (5 points)



5. A 40.0 kg child standing on a frozen pond throws a 0.5 kg stone to the east with a speed of 5.0 m/s. Neglecting friction between child and ice, find the recoil velocity of the child. (2 points)

6. A 5.00 g bullet moving with an initial speed of 400 m/s is fired into and passes through a 1.00 kg block, as shown in Figure. The block, initially at rest on a frictionless, horizontal surface, is connected to a spring of force constant 500 N/m. If the block moves 5.00 cm to the right after impact, find
- (a) the speed at which the bullet emerges from the block. (4 points)
- (b) the energy lost in the collision. (3 points)



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